User Manual for

Fast3VmrMLM

Fast **three-V**ariance-components **m**ulti-locus **r**andom-SNP-effect **M**ixed **L**inear **M**odel tools for genome-wide association study

(**version 2.0**)

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Disclaimer: The software has undergone comprehensive testing by Yuan-Ming Zhang's Lab at College of Plant Science and Technology, Huazhong Agricultural University. The results obtained from the software are generally reliable, correct, and appropriate. However, it is important to note that these results are not guaranteed for specific datasets. We strongly recommend that users integrate Fast3VmrMLM results with those from other software packages, i.e., 3VmrMLM and q3VmrMLM.

Download website:
https://github.com/YuanmingZhang65/Fast3VmrMLM

Citation:

Wang JT#, Chen Y#, Shu G#, Zhao M#, Zheng A, Chang X, Li G, Wang Y* and Zhang YM*. Fast3VmrMLM: A fast algorithm that integrates genome-wide scanning with machine learning to accelerate gene mining and breeding by design for polygenic traits in large-scale GWAS datasets. Plant Communications 2025; 6: 101385.

Wang JT#, Han XL#, Zhao MM, Zhang HQ, Chen Y, Jiang QY and Zhang YM*. A fast method for breeding by design via G×E interactions detected in large-scale climatic, phenomic and genomic data. National Science Review 2026, 13(10), nwag095.

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1 Introduction

1.1 Why Fast3VmrMLM?

Fast3VmrMLM (**Fast three-V**ariance-component multi-locus random-SNP-effect Mixed Linear Model) is an R package designed for identifying QTNs in a single dataset (one environmental dataset or the BLUP / mean data from multiple environmental datasets) and QTN-by-environment interactions (QEIs) in multi-environment joint analysis via fast, efficient, and big-data genome-wide association studies (GWAS). This method considers all possible genetic effects and controls for all possible polygenic backgrounds. These effects are replaced by an effect vector, while their polygenic backgrounds are replaced by a vector-based polygenic background vector. Thus, 5 and 10 variance components, respectively, in QTN and QEI detection are compressed into 3 variance components. This is named a compressed (three) variance component mixed linear model.

Fast3VmrMLM v2.0 with four modules is used to identify QTNs and QEIs via SNPs (Fast3VmrMLM), bin/gene haplotypes (Fast3VmrMLM-Hap), pedigree-inferred marker genotypes in crop breeding (Fast3VmrMLM-NCII), and other molecular markers (Fast3VmrMLM-mQTL):

1) Fast3VmrMLM: The genome-wide scanning plus machine learning framework was integrated with advanced computational and statistical techniques to identify QTNs and/or QEIs for complex traits using a fast, efficient, and big-data GWAS algorithm, Fast3VmrMLM.

2) Fast3VmrMLM-Hap: The Fast3VmrMLM-Hap module is used to associate the haplotypes of bins constructed by adjacent linkage disequilibrium markers with the trait of interest.

3) Fast3VmrMLM-NCII: The Fast3VmrMLM-NCII module is used to associate the marker genotypes inferred from parental marker genotypes with the trait of interest in the North Carolina Design II scheme or other mating designs (crop breeding populations).

4) Fast3VmrMLM-mQTL: To identify molecular QTLs (mQTLs), the Fast3VmrMLM-mQTL module can be used to associate the trait of interest with molecular variations (genes, lncRNA types, SVs, and k-mer). The input genotype file for molecular variations has two types of formats. One is the "*.vcf" format with multi-allele markers, and the other is PLINK binary genotype format with the start and end positions on the genome for molecular markers (see Table 6).

All of above three modules support single-environment GWAS (directly identifying QTNs and indirectly identifying QEIs/QMIs) and multi-environment joint GWAS (directly identifying QTNs and QEIs).

Fast3VmrMLM v2.0 works on the Windows and Linux systems.

1.2 Getting started

Fast3VmrMLM v2.0 (https://github.com/YuanmingZhang65/Fast3VmrMLM) is a package that runs in the R environment on the Linux system, which can be freely downloaded from the above website or requested from the maintainer, Dr Yuan-Ming Zhang at College of Plane Science and Technology, Huazhong Agricultural University (450625680@qq.com; soyzhang@mail.hzau.edu.cn).

1.2.1 Installation in the Linux

Within the Linux system, the Fast3VmrMLM software can be installed by the following four steps:

1) To download and install Miniconda3 at https://repo.anaconda.com/miniconda/, in which the recommended installer file is Miniconda3-py38_23.11.0-2-Linux-x86_64. Detailed for installing Miniconda3 were shown in §5.1.

2) To create a new conda environment named Fast3VmrMLM and install R version 4.3 by the following code:

conda create -n "Fast3VmrMLM" r-essentials r-base=4.3

conda activate Fast3VmrMLM

3) To install the dependency packages.

- Install dependency packages by following codes in Linux environment (or by *install.packages* in R environment):

  conda install -c conda-forge r-Rcpp=1.1.0 r-RcppArmadillo=15.0.2_1 r-data.table=1.15.2 r-RcppParallel=5.1.9 r-MASS=7.3_60.0.1 r-openxlsx=4.2.8 r-BH=1.87.0_1 r-BEDMatrix=2.0.4

4) Downloading Fast3VmrMLM from Github and installing it:

- Downloading via R code in R environment (R version 4.3):
- *download.file("https://codeload.github.com/YuanmingZhang65/Fast3VmrMLM/zip/refs/heads/main",destfile="Fast3VmrMLM-main.zip",mode="wb", method="libcurl")*

- Installing Fast3VmrMLM in R environment (R version 4.3):
  unzip("Fast3VmrMLM-main.zip")
  unzip("Fast3VmrMLM-main/Fast3VmrMLM_Linux_2.0.zip")
  install.packages("Fast3VmrMLM", repos = NULL)

Users can install and unzip ARM architecture Fast3VmrMLM in the R environment (R version 4.3) **if the server uses an ARM architecture CPU**:
  unzip("Fast3VmrMLM-main/Fast3VmrMLM_Linux_arm_2.0.zip")
  install.packages("Fast3VmrMLM", repos = NULL)

1.2.2 Installation in the Windows system

1) Download the installer of R 4.3.2 at https://cran.r-project.org/bin/windows/base/old/4.3.2/R-4.3.2-win.exe, and install R 4.3.2.

2) Download the installer of Rtools 4.3 at https://cran.r-project.org/bin/windows/Rtools/rtools43/files/rtools43-5976-5975.exe, and install R 4.3.2.

3) To install the dependency packages in R environment (R version 4.3.2):

- *pkgs=c("Rcpp_1.0.14","RcppArmadillo_14.0.2-1","RcppParallel_5.1.10","data.table_1.17.0","MASS_7.3-60.0.1","openxlsx_4.2.8","BH_1.87.0-1","stringi_1.8.7","zip_2.3.2","BEDMatrix_2.0.4","crochet_2.3.0")*
- *mirror=https://mirrors.tuna.tsinghua.edu.cn/CRAN/bin/windows/contrib/4.3/*
- *install.packages(paste0(mirror, pkgs, ".zip"), repos=NULL, type="win.binary")*

4) Downloading Fast3VmrMLM from Github and installing it:

- Downloading via R code in R environment (R version 4.3):
- *download.file("https://codeload.github.com/YuanmingZhang65/Fast3VmrMLM/zip/refs/heads/main",destfile="Fast3VmrMLM-main.zip",mode="wb", method="libcurl")*
- Installing Fast3VmrMLM in R environment (R version 4.3):
  unzip("Fast3VmrMLM-main.zip")
  install.packages("Fast3VmrMLM-main/Fast3VmrMLM_2.0.zip",repos=NULL,type="win.binary")

If you have any questions about installation or anything else, please don't hesitate to contact Prof. Yuan-Ming Zhang (soyzhang@mail.hzau.edu.cn; 450625680@qq.com) or Dr. Jing-Tian Wang (1390032282@qq.com).

User Manual file Users can decompress the Fast3VmrMLM package and find the User Manual file (name: Instruction.pdf) in the folder of "…/ Fast3VmrMLM/inst". Users also download it from the Fast3VmrMLM website.

Example datasets Users can decompress the Fast3VmrMLM package and find the example datasets in the folder of "…/Fast3VmrMLM/extdata".

1.2.3 Run Fast3VmrMLM

Once the software Fast3VmrMLM is installed, users may run it using two commands:

library(Fast3VmrMLM)

Fast3VmrMLM(***) (please see the example of § 2.2)

Before using Fast3VmrMLM in Linux, make sure to activate the conda environment of Fast3VmrMLM: *conda activate Fast3VmrMLM*. Users need to run *library(Fast3VmrMLM)* every time before utilizing the Fast3VmrMLM software package.

2 Function Fast3VmrMLM

Two main functions are provided: *Fast3VmrMLM()* for single-environment GWAS and *Fast3VmrMLM_MEJA()* for multi-environment joint GWAS. They share largely similar parameters; the parameters specific to *Fast3VmrMLM_MEJA()* are explicitly indicated in the table below.

2.1 Parameter settings

Parameter	Meaning	File format	Note
fileGen	Name & path of genotypic file in your device, e.g., fileGen="D:/Users/Genotype". **All file paths should exclude "~".**	PLINK binary files: **Genotype.bed+Genotype.bim+Genotype.fam**	
filePhe	Name & path of trait phenotypic file in your device, e.g., filePhe="D:/Users/Phenotype.csv".	*.csv (Phenotypic values. Row: individuals; Column: traits)	**Table 1**
fileKin	Name & path of individual kinship file in your device, e.g., fileGRM="D:/Users/GRM.csv" or fileGRM=NULL.	*.csv (GRM. Row & Column: individuals)	**Table 2**
filePS	Name & path of covariates file in your device, e.g., filePS="D:/Users/covariates.csv" or filePS=NULL.	*.csv (Population structure, or principal components, or evolutionary population structure. Row: individuals; Column: sub-populations / principal components / evolutional types or covariates, e.g., sex and age)	**Tables 3-5**
filePedigree	filePedigree="D:/Users/Pedigree.csv" when using the NC II genetic mating design population, and filePedigree=NULL (the default value) for all others.	*.csv (Pedigree of NCII population consists of 3 individual-ID columns: the female & male parents, and their F_1 offspring)	**Table 7**
PopStrType	The types of population structure include "PopStrType=Q" (*Q matrix or ancestry coefficient matrix calculated by Admixture software*), "PopStrType=PC" (*principal components or covariates*), and "PopStrType=Evol" (*evolutionary population structure*).		
fileOut	Save path of the result in your device, e.g., "D:/Users/".		
genoType	Setting the algorithm as "SNP" (SNP-based Fast3VmrMLM module), "Hap" (Haplotype-based Fast3VmrMLM-Hap module) and "molecular" (molecular-based Fast3VmrMLM-mQTL module)		
population	population="NCII" when using the NCII design, and population="nature" (the default value) for all others		
trait	Traits analyzed from number 1 to number 2, e.g., trait=1:3: users analyze the first to third traits.		
svrad	A physical distance of sliding window for removing potential candidate variants with the collinearity of the most significant one. Default value is svrad=2.0e+4 (bp). Users can obtain more potential associated variants by setting a small value of SearchRadius.		
svpal	A critical *P*-value (default svpal=1.0e-2) to select all potentially associated variants/loci in genome-wide single-variant scanning (the first step). The parameter svpal may be changed based on sample size see §2.3. The P-vale differs from 1.0e-5 for large population (>4000) to 1.0e-2 for small population.		

svmlod	A critical LOD score is set as svmlod=3 (default) for identifying suggested variants/loci, while a critical P-value is set as svmlod=0.05/m (Bonferroni correction; m: the number of markers) for identifyingg significant variants/loci.
SampleMarkersforGRM	A parameter indicates whether a subset of variants is used to construct kinship (TRUE) or not (FALSE). Default value is SampleMarkersforGRM=FALSE.
SampleMarkersforGRMNum	A parameter (>0) indicates the number of variants sampled to calculate kinship matrix. Only used when SampleMarkersforGRM=TRUE. Default value is SampleMarkersforGRMNum=1.0e+4.
c_threshold	A parameter for evaluating linkage disequilibrium of adjacent variants when constructing bin-based haplotypes, ranging from 0 to 1. Default value is c=0.7.
numofHaplotypes	A parameter for setting the number of haplotypes for each bin genotype in the Fast3VmrMLM-Hap and Fast3VmrMLM-mQTL module. Default value is numofHaplotypes=3.
nThreads	A parameter indicates the number of cores used in parallel computing. Default value is set as nThreads=20.
DrawPlot	A parameter indicates whether drawing and outputting the Manhattan plot based on the GWAS results. Default value is DrawPlot=FALSE.
Plotformat	A parameter indicates the format of the Manhattan plot. Default value is Plotformat=*.tiff.
MGinputClass	Setting the input file format "bed" (PLINK binary file) and "vcf" (.vcf format file), when variableType="molecular" in the Fast3VmrMLM-mQTL module. Default value is inputClass="vcf".
filegeneRegion	Setting the input file that indicates the genome intervals of each gene or molecular marker when inputClass="bed" in the Fast3VmrMLM-mQTL module.
n_en	A parameter for Fast3VmrMLM_MEJA function, a vector setting the environment number of each trait, etc., when there were two traits measured under three and four environments, respectively, setting n_en=c(3,4).
effect_out	effect_out=TRUE: Output all the effects of genome-wide scanning into the 'intermediate results' file, including additive and dominant effects in single-environment GWAS, and additive, dominant, additive×environment interaction and dominant×environment interaction effects in multi-environment GWAS. effect_out=FALSE: Do not output all the effects of genome-wide scanning into the 'intermediate results' file. Default value is effect_out=FALSE.

2.2 Running codes

2.2.1 Running codes for single-environment GWAS in R v4.3

The running code for Fast3VmrMLM module is as follows:

```
Fast3VmrMLM(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="SNP",trait=1,svrad=2e+4,svpal=1e-2,svmlod=3,nThreads=20,DrawPlot=FALSE, Plotformat="*.tiff")
```

To create a Manhattan plot, please set the parameter, *DrawPlot=TRUE*. This will output a Manhattan plot for each trait. Alternatively, users can draw the plot using the *Fast3VmrMLM_manplot()* function, as detailed in §4.1 in Instruction 2.0.

The running code of Fast3VmrMLM when using NCII design is as follows:

Fast3VmrMLM(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",filePedigree="/home/Pedigree.csv",PopStrType="PC",population="NCII",fileOut="/home/",genoType="SNP",trait=1,svrad=2e+4,svpal=1e-2,svmlod=3,nThreads=20,DrawPlot=FALSE,Plotformat=".tiff")*

When using partial NCII breeding population datasets via the Fast3VmrMLM module, the genotype file (*fileGen*) should have parental genotypes, and offspring's genotypes can be inferred from their parents by Fast3VmrMLM according to the pedigree (*filePedigree*).

The running code for Fast3VmrMLM-Hap module:

Fast3VmrMLM(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="Hap",trait=1,svrad=2e+4,svpal=1e-2,svmlod=3,c_threshold=0.7,numofHaplotypes=3)

The running code for Fast3VmrMLM-mQTL module:

Fast3VmrMLM(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="molecular",trait=1,svrad=2e+4,svpal=1e-2,svmlod=3,numofHaplotypes=6,MGinputClass="vcf")

Users must set "fileGen", "filePhe", "trait", and "fileOut", while the other parameters may be default in function Fast3VmrMLM (see § 2.1).

2.2.2 Running codes for multi-environment GWAS

The code for **multi-environment GWAS** is the same as that shown above, except for two changes: the function *Fast3VmrMLM()* should be replaced by *Fast3VmrMLM_MEJA()*, and the n_en parameter should be provided.

The running code for Fast3VmrMLM in multi-environment GWAS is as follows:

Fast3VmrMLM_MEJA(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="SNP",trait=1:2,n_en=c(3,2),svrad=2e+4,svpal=1e-2,svmlod=3,nThreads=20,DrawPl ot=FALSE,Plotformat="*.tiff")

The running code of Fast3VmrMLM module in multi-environment GWAS when using NCII dataset is as follows:

```
Fast3VmrMLM_MEJA(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",filePedigree="/home/Pedigree.csv",PopStrType="PC",population="NCII",fileOut="/home/",genoType="SNP",trait=1,n_en=c(2),svrad=2e+4,svpal=1e-2,svmlod=3,nThreads=20,DrawPlot=FALSE,Plotformat="*.tiff")
```

The running code for Fast3VmrMLM-Hap module:

```
Fast3VmrMLM_MEJA(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="Hap",trait=1,n_en=c(2),svrad=2e+4,svpal=1e-2,svmlod=3,c_threshold=0.7,numofHaplotypes=3)
```

The running code for Fast3VmrMLM-mQTL module in multi-environment GWAS:

```
Fast3VmrMLM_MEJA_SV(fileGen="/home/Genotype.vcf",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",trait=1:2,n_env=c(2, 2),svrad=2e+4,svpal=1e-2,svmlod=3)
```

2.3 Parameter adjustment

When applying the new method, three parameters may be adjusted, including svpal, c_threshold and numofHaplotypes. First, the svpal was used to set the P-value threshold for potential QTNs/QEIs in the first step. The recommended threshold depends on the population size: the larger the population size, the smaller the threshold. For example, svpal = 0.05 or 0.01 for a population size of less than 4,000, and svpal = 1e–5 for a population size of more than 20,000. Second, the c_threshold and numOfHaplotypes parameters can be adjusted when constructing haplotypes in Fast3VmrMLM-Hap. c_threshold (0 to 1; 0.7 is recommended) is used to determine bin markers: the smaller the value, the larger the number of bin markers. The numofHaplotypes parameter determines the maximum number of haplotypes (e.g., three).

2.4 Data input format

Format for genotypic dataset “fileGen”

The file type of genotypes is "plink binary format" (Genotype.bed + Genotype.bim + Genotype.fam), which can be found the introduction from PLINK v1.9 (https://www.cog-genomics.org/plink2/.

Format for genotypic dataset "filePhe" (Table 1.1 and Table 2.2)

The file type of phenotypes for complex trait is *.csv, following the format outlined in Table 1.1. The first row in the first column: "<Phenotype>"; the second to *n*th rows in the first column: individual IDs or names, such as Ind46. The first row in other columns: trait names, such as "trait1", and the second to *n*th rows in other columns: values of phenotypic traits. The phenotypes missed: "NA".

The phenotype file format for **multi-environment joint GWAS** is shown in Table 1.2, where each column represents one trait measured under one environment. For example, Table 1.2 includes two traits (trait1 and trait2) measured across two and three environments, respectively, and the n_en parameter in *Fast3VmrMLM_MEJA()* function should be set as: n_en=c(2,3)

Table 1.1. The format of phenotypic dataset

<Phenotype>	**trait1**	**trait2**	**trait3**	**…**
Ind46	91.03	88.32	87.67	…
Ind52	103.11	103.10	98.36	…
Ind57	92.07	116.30	NA	…
Ind64	128.5	101.20	101.02	…
Ind68	95.84	91.74	94.50	…
⋮	⋮	⋮	⋮	…

Table 1.2. The format of phenotypic dataset for multi-environment GWAS

<Phenotype>	**trait1_env1**	**trait1_env2**	**trait2_env1**	**trait2_env3**	**trait2_env3**	**…**
Ind46	91.03	88.32	116.30	NA	87.67	…
Ind52	103.11	103.10	101.20	103.11	98.36	…
Ind57	92.07	116.30	92.07	87.67	NA	…
Ind64	128.5	101.20	128.5	103.10	101.02	…
⋮	⋮	⋮	⋮	⋮	⋮	…

Format for genotypic dataset "fileKin" (Table 2)

The "fileKin" should be a file with *.csv format. All the kinship coefficients are listed as an n × n matrix. Both rows and columns represent individuals that arranged in the order of the *.fam file. The parameter “fileKin=NULL” indicates that the kinship matrix is calculated by the “Fast3VmrMLM” software. When fileKin="D:/Users/kinship.csv", the kinship matrix with name kinship.csv is uploaded from the folder "D:/Users". If the number and order of individuals in the "kinship.csv" file do not match those in the phenotypic files, our software will match them.

Table 2. The format of knship dataset

Ind46	Ind52	Ind12	Ind143	Ind57	Ind67
1	0.700361011	0.599277978	0.675090253	0.620938628	…
0.700361011	1	0.620938628	0.666064982	0.653429603	…
0.599277978	0.620938628	1	0.561371841	0.5433213	…
0.675090253	0.666064982	0.561371841	1	0.615523466	…
0.620938628	0.653429603	0.5433213	0.615523466	1	…
⋮	⋮	⋮	⋮	⋮	…

Q matrix format for dataset “filePS” (Table 3)

The Q matrix dataset in Table 3 consists of a (n+1) × (k+1) matrix, where n is sample size (the number of the above common individuals), and k is the number of sub-populations. The first column is “<Structure>” and individual IDs or names. In the 2nd to (k+1)-th columns, “Q1” to “Qk” indicate sub-populations. In the second row, “0.014”, “0.972” and “0.014” are posterior probabilities that the individual “33-16” is belong to the 1st, 2nd, and 3rd subpopulations, respectively.

Table 3. The Q matrix format of dataset filePS

<Structure>	Q1	Q2	Q3
Ind46	0.014	0.972	0.014
Ind52	0.003	0.993	0.004
Ind57	0.071	0.917	0.012
Ind64	0.035	0.854	0.111
⋮	⋮	⋮	⋮

Principal components or covariates for dataset “filePS” (Table 4)

The principal components or covariates dataset in Table 4 consists of a (n+1) × k matrix, where n is sample size (the number of the above common individuals), and k is the number of principal components and/or covariates.

The first column is "<PCA>" and individual IDs or names. The 2nd to *k*-th columns indicate principal components and/or covariates.

Table 4. The principal components or covariates format of dataset filePS

<PCA>	PC1	PC2	PC3	…
Ind46	0.306	0.029	0.226	…
Ind52	-0.708	-0.271	1.413	…
Ind57	-2.330	0.116	-0.824	…
Ind64	1.059	0.470	-0.135	…
⋮	⋮	⋮	⋮	

The evolutionary population structure for dataset "filePS" (Table 5)

The evolutionary population structure dataset in Table 5 consists of a $(n+1) \times 2$ matrix, where n is sample size and the number of categories for variables is the number of sub-populations. The first column is "<EvolPopStr>" and individual IDs or names. The 2nd column indicates the evolutionary sub-population. Other population structure described by character type of variables are supported in this format.

Table 5. The evolutionary population structure format of dataset filePS

<EvolPopStr>	EvolType
Ind46	A
Ind52	B
Ind57	A
Ind64	C
⋮	⋮

The format of the chromosome intervals of each molecular marker for the parameter filegeneRegion in § 2.1 (Table 6)

When genoType="molecular" and MGinputClass="bed", the genome intervals of each molecular marker should be set by users via the file in Table 6. The first column is its ID, the second column is its chromosome, the third and the fourth columns are the left and right physical positions, respectively.

Table 6. The chromosome intervals of each molecular marker

Zm00001eb000010	1	34617	40204
Zm00001eb000020	1	41214	46762
⋮	⋮	⋮	⋮

The format of the pedigree file in § 2.1 (Table 7)

When population="NCII", the filePedigree should be set according to Table 7. It contains three individual ID columns: the female parent (first column), the male parent (second column), and their F1 offspring (third column).

Table 7. The pedigree file

LT16	LT131	LT16_LT131
LT07	LT131	LT07_LT131
⋮	⋮	⋮

2.5 Result

The results include three files: *_intermediate.csv (intermediate results), *_result.csv (final results) and a Manhattan plot file.

*_intermediate.csv: This file contains the results of genome-wide single-variants scanning in the first step. In this file, all the columns are named as "MarkerID" (variants name), "CHR" (Chromosome), "POS" (variants position (bp) on the genome), and "pval" (the *P*-value for main-effect variants).

MarkerID	CHR	POS	pval
PZB00859.1	1	157104	0.292043111
PZA01271.1	1	1947984	0.185246808
PZA03613.2	1	2914066	0.99208603
PZA03613.1	1	2914171	0.999987108
⋮	⋮	⋮	⋮

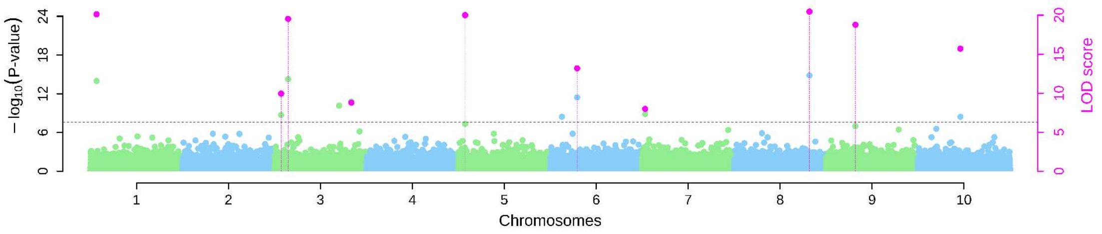
*_result.xlsx: The final result file for the Fast3VmrMLM module is as follows. All the columns are named as "Marker" (marker name); "Chromosome"; "Position (bp)" (markers position (bp) on the genome); "add" (additive effect); "dom" (dominance effect); "LOD" (LOD score); variance (the variance of each variants); "r2(%)" (the proportion of total liability variance explained by each variants); "*P*-value" (calculated from LOD score using χ^2 distribution); "significance" (significant (SIG) variants are based on Bonferroni correction, that is, critical *P*-value is $0.05/m$, where m is the number of tests or variants, while suggested (SUG) variants are based on LOD ≥ 3.0, default). "A allele" is the nucleotide base with a dummy variable of 1.0.

Trait ID	Trait name	Marker	Chromo	Position (bp)	LOD	add	dom	variance	r^2(%)	P-value	signific	A allele
1	phe_1	null_653	1	654	31.5242	-0.4295	5.6711	7.9444	5.0209	3.00e-32	SIG	C
1	phe_1	null_20728	3	20729	9.9457	1.6284	2.7827	4.3464	2.7469	1.13e-10	SIG	G
1	phe_1	null_52928	6	52929	13.1912	3.5793	-0.163	6.4797	4.0952	6.45e-14	SIG	A
1	phe_1	null_78191	8	78192	65.6504	6.5965	-1.579	5.054	3.1942	2.25e-66	SIG	T

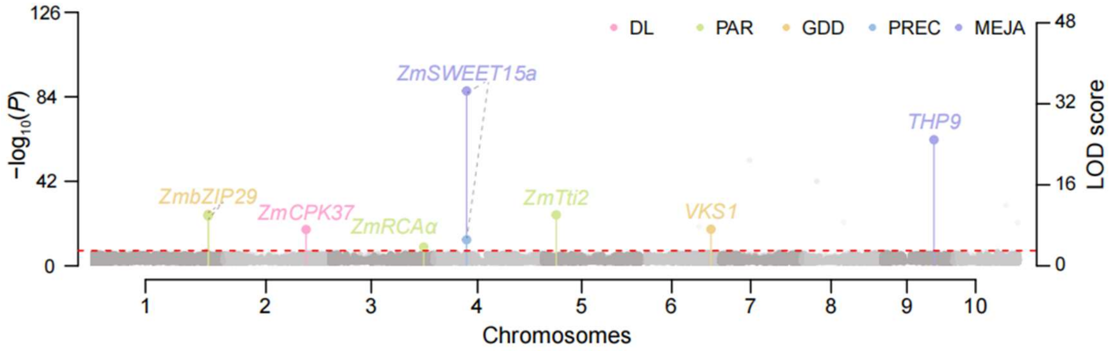
The final result file for the Fast3VmrMLM-Hap module is as follows. The "Marker_L" and "Marker_R" mean the markers on the left and right boundaries, respectively, of chromosome region (bin) determined by linkage disequilibrium, while "Position_L (bp)" and "Position_R (bp)" are for the left and right genomic positions, respectively. "Hap_1" to "Hap_numHap" are the haplotypic effects of the 1st to numHap-th haplotypes, where "numHap" is No. of haplotypes.

Trait	Trait	Marker_L	Marker_R	Chr	Position_L (bp)	Position_R (bp)	LOD	Hap_1	Hap_2	Hap_3	numHap	variance	r^2(%)	P-value	significance
1	phe_1	rs16857692	rs4668204	2	170672831	170705429	5.78	0.018	-0.870	0.018	3	0.274	0.1268	1.63e-06	SUG
1	phe_1	rs10439267	rs6435173	2	203631692	203721357	125.53	-2.213	-2.595	-2.213	3	12.674	5.9419	2.99e-126	SIG
1	phe_1	rs1978865	rs7651321	3	61127327	61256881	82.85	-1.924	-1.616	-1.924	3	6.88	3.2254	1.42e-83	SIG

*_Manhattan plot: Y-axis on the left-side reports -log10 *P*-values of variants, which are obtained from single-variant genome-wide scanning for all the variants in the first step of Fast3VmrMLM, while Y-axis on the right-side reports LOD scores, which are obtained from likelihood ratio test for suggested and significant variants, with the suggested threshold of LOD = 3.0 (dashed line), in the second step of Fast3VmrMLM. Users can set different LOD threshold by setting svmlod (see § 2.1). These LOD scores are shown in points with straight lines. If LOD score ≥ 20, the LOD scores obtained are transformed as LOD' = 20 + (LOD – 20)/100 in order that the Manhattan plot is more beautiful.



We recommend marking all the all the significant and suggested QTNs with known and candidate genes in the Manhattan plot and listing all the significantly and suggested associated markers with the traits of interest in a supplemental table. For example, the Manhattan plot below depicts all the known and candidate gene-by-environment interactions, along with their corresponding QEIs/QMIs, as shown in Figure S2 of Wang et al. (2026).



3 References

[1] Wang JT#, Chen Y#, Shu GP#, Zhao MM#, Zheng A, Chang XY, Li GQ, Wang YB*, Zhang YM*. Fast3VmrMLM: A fast algorithm that integrates genome-wide scanning with machine learning to accelerate gene mining and breeding by design for polygenic traits in large-scale GWAS datasets. Plant Communications 2025; 6: 101385.

[2] Wang JT#, Han XL#, Zhao MM, Zhang HQ, Chen Y, Jiang QY and Zhang YM*. A fast method for breeding by design via G×E interactions detected in large-scale climatic, phenomic and genomic data. National Science Review 2026, 13(10), nwag095.

4 Additional functions in Fast3VmrMLM for pre- and post-GWAS analyses

There were some functions that can be called by users in R environment for pre- and post-GWAS analyses in Fast3VmrMLM software.

4.1 Draw a Manhattan plot

Users can draw a Manhattan plot by setting the parameter, *DrawPlot=TRUE*, when running the *Fast3VmrMLM()* and *Fast3VmrMLM_MEJA()* functions. Alternatively, the flexible *Fast3VmrMLM_manplot()* function can be used to adjust the coordinate axis range and produce a lightweight PDF output. This function can be accessed via the code below:

```
Fast3VmrMLM_manplot(midresult_dir="/home/trait1_intermediate.csv",
result_dir="/home/trait1_result.xlsx",fileOut=NULL,Plotformat="*.tiff",col
_background=c("lightgreen","lightskyblue"),adj_thres=150,log_times=5,
LOD_times=3,reduction=0.5)
```

Parameter description:	
midresult_dir	The intermediate result file output from GWAS (see § 2.2.1).
result_dir	The final result file output from GWAS (see § 2.2.1).
fileOut	The output address of Manhattan plot.
Plotformat	The formate of output Manhattan plot. Supporting the formate of tiff, pdf, .png, and jpeg.
col_background	The color of the points.
adj_thres	The threshold for adjusting LOD score. When LOD score is larger than *adj_thres* the following adjustment was used to replace it: LOD'=(LOD-*adj_thre*)/100 + *adj_thre*.
log_times	The maximum value of the left y-axis is set to *log_times* times the largest log *P-value*.
LOD_times	The maximum value of the right y-axis is set to *LOD_times* times the largest LOD score.
reduction	For large marker number (*m*), only a proportion (*reduction*×*m*) of markers is randomly sampled for plotting, which effectively reduces the size of the PDF Manhattan plot.

4.2 For individual-haplotype file

For the trait-associated haplotypes identified by Fast3VmrMLM-Hap, the individual–haplotype file records the haplotype of each individuals. It can be generated using *Fast3VmrMLM_Hapout()* function via the following code:

```
Fast3VmrMLM_Hapout(plinkFile="/home/gen",filePhe="/home/phe.csv",file
PS=NULL,fileOut=NULL,file_finalresult=NULL,c_threshold=0.7,numofHaplo
types=3)
```

Parameter description:	
file_finalresult	The final result file output from Fast3VmrMLM-Hap method (see § 2.2.1).
others	Other parameter should be same to the runing of Fast3VmrMLM-Hap.

Return value	A file including all individuals and all haplotypes in the *file_finalresult* file. where "1" indicates the presence of the haplotype in the individual.

4.3 For inferring the genotypes of offspring via pedigree and parents

The genotypes of offspring can be inferred from the pedigree and parental genotypes using the *Fast3VmrMLM_outcode()* function via the following code. Before using this function, a dependency R packages, BEDMatrix should be installed in R environment via *install.packages("BEDMatrix")*.
The genotypes of offsprings can be inferred from the pedigree and parental genotypes using the Fast3VmrMLM_outcode() function via the following code. Before using this function, the *BEDMatrix* R package dependency should be installed in the R environment via *install.packages("BEDMatrix")*.

Fast3VmrMLM_outcode(plinkFile="/home/gen", filePedigree=NULL)

Parameter description:	
plinkFile	The plink formate genotype file of parents.
filePedigree	The pedigree file (see Table 7).

Return value	A numerical matrix including all parents and offspring, where columns represent markers and rows represent individuals. Row names correspond to individual IDs.

5 Other instructions and solutions of Issues

1) Detailed for installing Miniconda3

First, downdloading the installer of Miniconda3:

wget https://repo.anaconda.com/miniconda/Miniconda3-py38_23.11.0-2-Linux-x86_64.sh

Then, installing Miniconda3:

bash Miniconda3-py38_23.11.0-2-Linux-x86_64.sh

yes

yes

source ~/.bashrc

Third, confirm the installation successful:

conda --version

If displayed "*conda 23.11.0*", the installation was successful.

2) Issue

There were “libstdc++.so.6: version `CXXABI_1.3.15' not found” reported after *library(Fast3VmrMLM)*.

> **Solution:** Firstly, confirm that the R version is 4.3 and try to run Fast3VmrMLM again. If it doesn’t work, update the version of libstdc++ by the following code:
>
> *cd /home/user/anaconda3/envs/Fast3VmrMLM/lib*
>
> *rm libstdc++.so*
>
> *rm libstdc++.so.6*
>
> *ln -s libstdc++.so.6.0.34 libstdc++.so*
>
> *ln -s libstdc++.so.6.0.34 libstdc++.so.6*